Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / bending
Case	rca_11c87937f7d845f288464edae46f2422
Date	2026-11-02 15:24:35
Status	AI proposal, pending expert review

1. Problem

Defect	Severity	Confidence	Detail
crack	critical	88%	
crack	critical	86%	
dent	major	82%	
dent	major	79%	
crack	critical	77%	
dent	major	51%	
dent	major	50%	
dent	major	17%	
scratch	minor	16%	
dent	major	11%	

2. Root cause

Cause: Material Hardness Variation

Reason: The critical cracks detected on the SPCC steel parts are directly linked to hardness variation in incoming material, which increases susceptibility to cracking during bending operations as described in [02cf0928-6739-4f7e-ae1b-f6740283f00f].

Effect: Cracking at bend radius

Confidence: 77%

Supporting evidence
- **02cf0928-6739-4f7e-ae1b-f6740283f00f**: Material Inconsistency / Hardness Variation Root Cause Analysis Defect Types: crack, deformation, dimensional variation Material: SPCC steel, cold-rolled sheet metal Root Cause: Hardness variation in incoming material lot causes unpredictable springback behavior and susceptibility to cracking under forming operations. Effect: Cracking at bend radius, dimensional inaccuracy due to variable springba

3. Solution

Corrective actions (fix this occurrence)

1. Verify material certificates against specification for yield and tensile strength
2. Conduct incoming material hardness testing

Preventive actions (stop it recurring)

1. Implement stricter supplier quality controls for incoming SPCC steel
2. Establish a material batch tracking system

Precautions

- Monitor for batch-to-batch variation in forming performance

How to verify the fix

Perform bend tests on samples from new material lots

4. Other causes considered

Rank	Cause	Confidence	Reason
2	Tooling Damage or Die Surface Degradation	71%	The major dents detected on the parts are consistent with damage to die surfaces, which causes surface defects during the bending process as identified in [587aa665-c6f1-48df-8783-09d65ff4f82a].
3	Springback	36%	While the primary defects are cracks and dents, springback is a common root cause in bending processes that can contribute to overall part deformation and dimensional instability, as noted in [6872e1ff-9d0f-41f8-9cd9-c783d66c1b46].

5. Traceability

AI model: gemini:gemini-3.1-flash-lite. Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.